

Multi Map Visual Localization for Unmanned Aerial Vehicles

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Abstract—Localization has long been an essential area of research within robotics. The popularity of using Unmanned Aerial Vehicles (UAVs) to solve different tasks has increased and is expected to continue. Developing a robust complementary system to the Global Navigation Satellite Systems (GNSS) used today has been researched, and visual localization using cameras and satellite images is a popular choice to use. One of the challenges with using satellite images is that different images over the same area can impact the system's performance. This article proposes a novel approach called Multi Map Visual Localization (MMVL), a method to use multiple satellite images simultaneously, which is combined using a weighted average of probability maps. The proposal uses a convolutional neural network (CNN) with a caching strategy together with Monto Carlo Localization (MCL). MMVL achieves excellent robustness compared to other approaches and manages to estimate the correct location on all test flights. At the same time, using multiple satellite images does not significantly impact accuracy and computation time.

Index Terms—Localization, aerial systems: perception and autonomy, computer vision for automation.

I. INTRODUCTION

ACCURATE and robust localization has been considered an important topic in robotics for a long time. Some have even called it the “most important problem in robotics” [1]. Different types of robots require different types of localization depending on the problem they try to solve and their limitations. UAVs, is a rising field of robotics where the industry has gone from a military asset to a billion-dollar industry [2]. They are now used to solve problems like aerial photography, geographic mapping, disaster management, agriculture surveillance, and aerial inspection [3]. The growth in the UAV market is expected to continue as technology is maturing [3], [4]. One example is the delivery industry, where big companies like Amazon are developing systems to deliver packages directly to consumers [5].

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For many of these use cases, robust and accurate localization methods are essential to succeed in the task and keep safety high.

Today, the primary localization method is with GNSS. There are a few different systems, but generally, they all work in the same way. The most famous is the American GPS. These systems work by receiving signals from multiple satellites in orbit and calculating a position based on the time the signals used to get to the receiver [6]. As long as the receiver has an unobstructed view of the sky, GNSS works everywhere and is very accurate with errors typically down to a couple of meters [7]. However, GNSS require a good signal from the satellites to work. The signals can be impacted by natural causes such as occlusion, multi-path reception, or none-line-of-sight (NLOS) [8]. In addition, humans can destroy or alter the signals using jamming and spoofing [6]. This is primarily military technology, but the impact on civilian sector is also increasing [9].

Due to the downside of GNSS, work has been done to find an alternative localization method. One promising approach is the use of visual localization. Here, a UAV camera gathers information about the environment and compares it with a reference map. The benefit of using a camera is that it is cheap, information-rich, and already included in most of the UAV systems. Different reference maps have been used, but using satellite images is tempting, as they contain a lot of information and are readily available. Through sites like Google Earth, it is even possible to specify the date of the satellite image. The challenge is to develop a method that works in all situations. The differences between satellite and UAV images, and between different satellite images, can be significant, as they are captured at different times, lighting conditions, weather, etc. Methods might work well with one satellite image but not another taken at a different time. In addition, the large amount of data in images makes it difficult to create efficient strategies to be deployed on small UAVs.

The goal of this letter is not to achieve better accuracy than modern GNSS but rather to develop a complementary method for the situations when GNSS is not working. We propose a novel method, called MMVL, that uses multiple satellite images simultaneously. This is designed to increase the system's robustness as it is impossible to know which satellite image the localization will work on before flight. MMVL uses images taken straight down to localize the UAV in two dimensions, (x, y) . The system is built around MCL using a CNN to compare images. The satellite maps are combined using a weighted average, and a probability map is created at each iteration to estimate the most

TABLE I
PRIOR APPROACHES

	abBRIEF	NBD-BRIEF	Samano	SIVL	LSVL
Authors	Mantelli et al. [10]	Stein et al. [11]	Samano et al. [12]	Kinnari et al. [13]	Kinnari et al. [14]
Map type	Satellite	Building footprint	Cartographic	Satellite	Satellite
Localization method	MCL	MCL	MCL	MCL	Point mass filter
Image comparison	BRIEF	BRIEF	resnet18/resnet 50	6-layer CNN	resnet50+CapsNet
Caching	No	No	Yes	No	Yes
Real flights	Yes	Yes	No	Yes	Yes
Preprocessing	Color space & quantization	Segmentation & proximity	No	Orthorectification	Orthorectification
Altitude (m)	35 – 300	35 – 180	-	91, 92	54, 71, 84, 91
Error (m)	17.8	9.1 – 57.6	< 95	26.5 – 30.6	12.6 – 18.7

likely positions. A cache is used to increase the computation efficiency. MMVL was tested on four different flights. The main contributions of this letter are:

- Using multiple satellite images as reference maps simultaneously can increase the system's robustness without decreasing the accuracy.
- Proposing a novel strategy to combine multiple satellite images using a weighted average, where the weight is based on the normalization of the scores for each map.
- When using neural networks, caching strategies are necessary to increase the computation efficiency.

II. RELATED WORK

Many methods have been proposed to solve the problem and achieve robust and accurate visual localization. This work will focus on absolute visual localization (AVL). This means localization in the world coordinates without prior knowledge using cameras as the sensor. This differentiates it from relative methods like visual odometry (VO), where the location found is relative to the start position. Table I shows some prior proposals, and the following is a short presentation.

MCL is a popular algorithm proven to work in AVL multiple times. MCL is a probabilistic localization method proposed by Dellaert et al. in 1999 [15]. The method models the probability distribution $p(\mathbf{x}_k|Z^t)$ using particles. Where $\mathbf{x}_t$ is possible locations, and Z_t is all measurement, at time t . This is estimated recursively using a *prediction model* and a *measuring model*. The algorithm has been successfully used in many problems with different robots.

A common approach has been to use image descriptors to compare images. One of the more successful methods was the abBRIEF proposed by Mantelli et al. [10]. It was proposed as a method that is robust to different lighting conditions in the images using a modified version of the BRIEF descriptor. The images were transformed to the CIELAB color space, and the a^* and b^* channels were used. In addition, a quantization step was applied to smooth out minor differences. The descriptors were used to compare images to be used with MCL. The proposed method achieved high robustness to different conditions and has become popular as a baseline.

With the increase in popularity of neural networks and especially CNNs in the 2010 s, more and more problems have been solved using machine learning, helped by popular models like VGG [16] and ResNet [17]. However, as noted by Couturier and Akhloufi in 2021 [18], the number of AVL proposals

using CNNs was few. The authors mentioned difficulties in creating end-to-end architectures and meeting the computational demands such systems require. However, some methods have been proposed using neural networks to different degrees. In recent years, it has become more popular.

Some methods have successfully used the segmentation of buildings in the images using the U-net architecture [19]. Ovstegard et al. [20] used building information to create a binary descriptor. Choi and Myung [21] used the ratio of buildings in the image, and Stein et al. [11] proposed a modified BRIEF descriptor, called NBD-BRIEF, using the distance from the buildings to manage more complex situations. All approaches work but require buildings to be present in the images, and they are using building footprint maps instead of satellite images as reference maps.

Samano et al. [12] proposed a method using MCL and similarity learning, a concept in machine learning first proposed by Bromley et al. [22]. The method used a ResNet18 model to create embeddings from a cartographic map and compared with embeddings from the UAV images created by a ResNet50 model. The embedding was compared using Euclidean distance and then fed to the MCL algorithm. Embeddings from the map were cached beforehand to decrease the computational requirement under flight. This was done by splitting the map into a grid. The score of the particles was found by using interpolation of the eight spatially closest embedding. The proposed model achieved quick convergences but was only tested on simulated data. The method also requires the availability of a cartographic map of the area.

Kinnari et al. proposed two methods, using CNNs and similarity learning. The first proposal is called Season Invariant Visual Localization (SIVL) [13]. This used a custom 6-layer CNN to create image embeddings with two fully connected layers for comparing them. Before the model, the orthorectification process from their earlier work [23] was applied to the images from the UAVs. The orthorectification changes the perspective of an image, from an image taken at an angle to an image taken orthogonal to the ground. For the localization, MCL was used. The results show that the method works on simulated and real flights at a fixed altitude and simulated VO.

The second proposal from Kinnari et al. [14] was a method they called Large-Scale Visual Localization (LSVL). The model applied the same orthorectification on the UAV images as prior proposals [23]. Two different CNN architectures were proposed; both used ResNet50, but the first used a single fully connected layer in addition, and the second used a CapsNet [24]. The model

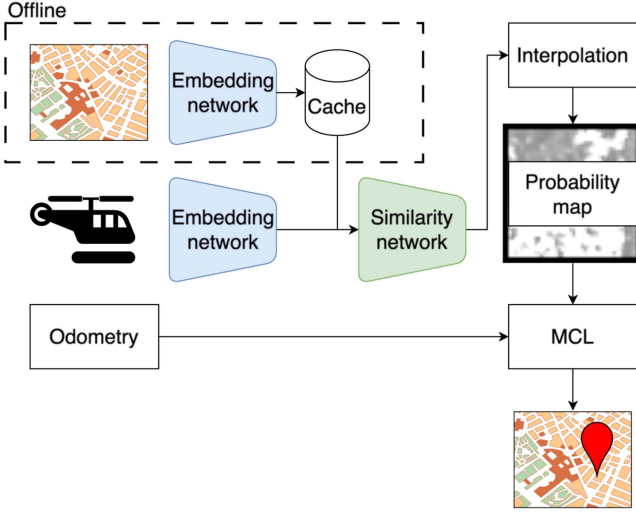


Fig. 1. Overview of the different stages and parts of MMVL.

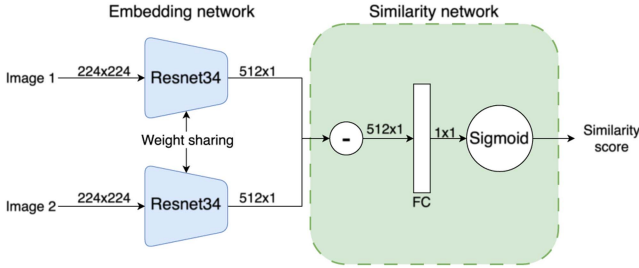


Fig. 2. The architecture of the Similarity model which consists of an Embedding network and a Similarity network.

used a similar caching strategy as Samano et al. [12]. The authors chose not to use MCL, but rather a method they called point mass filter. The tests again had a fixed height per flight and used simulated odometry data, but the results from the tests achieved very high accuracy.

As described in this section and by Table I, many different proposals for visual localization achieve good results. However, the problem of visual localization has not been solved. Different satellite images over the same area can significantly impact the methods' performance, but apart from abBRIEF [10], little work has been focusing on this. No prior method has used the multiple satellite images available to the method's advantage.

III. PROPOSED METHOD

MMVL is building upon prior work comparing images from the UAV with satellite images. Fig. 1 shows an overview of the proposed approach. The *embedding network*, a CNN, creates an embedding, a compact representation of an input image. The embeddings from the satellite image are cached offline to increase the computation speed. A small one-layer *similarity network* compares two embeddings and outputs a similarity score. Together, the embedding and similarity networks form the *similarity model*, shown in Fig. 2. A weighted average over all the different satellite maps is then used to create a probability

map inspired by grid-based localization [25]. The probability map estimates the *measuring model* used in MCL to find the localization of the drone in two dimensions, (x, y) . MMVL was designed to only need a camera and a cheap digital compass to work and be independent of altitude. Following is an explanation of the different parts¹.

A. Similarity Model

The embedding network for our proposal uses a pre-trained ResNet34 [17] model, fine-tuned with a custom dataset explained in Section IV-A. The last layer of ResNet34 was removed to output embeddings with 512 values.

The architecture of the similarity network is based on the proposal of Koch et al. [26], used to compare letters. The network is a single fully connected layer with 512 inputs, one output, and a sigmoid function, σ , as the activation function. The input to the network is the element-wise difference between two embeddings. (1) shows a mathematical expression of the similarity network.

$$s = \sigma \left(\sum_{j=1}^{512} w_j \left| \mathbf{e}_1^{(j)} - \mathbf{e}_2^{(j)} \right| \right) \quad (1)$$

Where w_j is the weights of the layer and $\mathbf{e}_1, \mathbf{e}_2$ are the embeddings of the two images that will be compared. s is the similarity between the two images as a number between 0 and 1. Details about the training are given in section IV-A.

B. Caching

A cache of the embeddings from the satellite maps is used to decrease the computational requirement during runtime of MMVL. This is similar to Samano et al. [12] and Kinnari et al. [14], where patches from the satellite images are processed offline before the flight. Patches with the size of 275×275 pixels from the satellite maps are used to create image embeddings. The embeddings were saved to create a cache with B embeddings. Different separation sizes between the patches' centers were tested, but the impact was minor. In this work, a separation of 100pixels was therefore used. The cache also enables the simultaneous usage of N maps. In that case, the cache is defined as $E = \{\mathbf{e}^{m,b}, m = 1..N, b = 1..B\}$, where m is the specific map and b is the index of the embedding.

Since the similarity model is trained to be altitude-independent, only one scale of each map is needed. Consequently, MMVL can not estimate the altitude, but it allows the cache to be small and efficient. For the flights used in this letter (explained in section IV-B), this resulted in a cache of 1936 embeddings for each satellite map, requiring only 4 MB of memory.

C. Probability Map

Using the cache and interpolation, a probability map of the same size as the satellite map is created at each timestep.

¹Code available at <https://github.com/tobiaslo/MMVL>

The probability map estimates the *measuring model*. At each timestep, t , the image from the UAV is rotated based on the drone's orientation and compared with all embeddings $\mathbf{e}^{m,b}$ in the cache E to get the score $s_t^{m,b}$ for every embedding. Calculating scores for all embeddings in the cache was needed to combine the different satellite images and will be explained later in this section. Furthermore, since the similarity model was small, calculating the scores from the embedding was efficient. More information about the computing times will be presented in section V-C.

It was found that transforming the scores with the function $\min(s_t + \lambda, 1)$, where $\lambda = 0.45$, helped MMVL to converge correctly. This is because errors in the similarity model were mitigated by increasing the probability that particles with low scores were sampled and treating high scores equally. The downside of this transform is a slower convergence, which will be shown in Section V.

A weighted average is used to combine the scores from the different maps. The scores, $s_t^{m,b}$, are transformed and normalized based on the scores from map m . The results are then averaged over every map. The normalization works as a weight, creating a weighted average. Scores from maps with many high similarities will be weighted lower than scores from maps with few high similarities. The equation for the combined score, s_t^b , can be written as (2).

$$s_t^b = \left(\sum_{m=1}^N \frac{\min(s_t^{m,b} + \lambda, 1)}{\sum_{k=1}^B \min(s_t^{m,k} + \lambda, 1)} \right) / N \quad (2)$$

The transformed similarity scores are placed in a grid according to the position from which the satellite patches were extracted. The grid is resized to the same size as the original satellite map using bicubic interpolation. This creates a likelihood for all positions $\mathcal{X}$ given the measurements $\mathcal{Z}$ and satellite map $\mathcal{M}$, written as $P(\mathcal{X}|\mathcal{Z}, \mathcal{M})$. The probability map is an estimation of this and models the measuring model in the MCL algorithm.

D. Prediction Model

The *prediction model*, also written as $p(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{u}_{t-1})$, is used in MCL to predict the next position of the UAV based on the movement. The movement, $\mathbf{u}$, is estimated using VO. The implemented method was used by both Mantelli et al. [10] and Stein et al. [11], and is a simple VO method. It uses SURF descriptors [27] and matching using FLANN [28] to calculate a relative motion for x, y , scale z and rotation θ . Our proposal only estimates the UAV in x and y coordinates, so an absolute rotation of the UAV is needed. As the dataset was collected before the project, this data needed to be simulated. The simulation uses the rotation from the ground truth, collected with GNSS, and an added noise of ± 5 degrees. Noise is set after typical errors of cheap magnetometer sensors.

Because of error and noise in the estimated movement, $\mathbf{u}_t$, uncertainty is modeled using a probability distribution. The distribution uses a Gaussian distribution with the variance proportional to the movement calculated, with a minimum variance of $\sigma_{\min}^2$ pixels. This is used on the assumption that the larger the movement, the higher the uncertainty becomes. For MMVL

$\sigma_{\min}^2 = 10$ was used. The prediction model can be written as

$$p(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{u}_{t-1}) = \mathcal{N}(\mathbf{x}_{t-1} + \mathbf{u}_t, \max(\sigma_{\min}^2, \mathbf{u}_t)) \quad (3)$$

IV. EXPERIMENTS SETUP

This section will go through the dataset collection and the training of the similarity model. Then, the datasets containing real-world UAV flights used to test MMVL will be presented. The section will end with a test of the accuracy of the similarity model.

A. Model Training

To train the model, a dataset of satellite images was collected from Google Earth. Coordinates from 116 Brazilian cities were found from simplemaps.com.² This includes cities from different regions and sizes. The satellite images were extracted with the coordinates in six different years, from 2018 to 2023. This means there are satellite images from one place in each city, but it has six different versions. Each image and version was split into patches of 250×250 pixels with no overlap between the patches. After cleaning and removing patches with minimal variance, like patches containing only clouds, the total number of images was 78331.

The embedding and similarity networks were trained simultaneously using Binary cross-entropy loss. The input images were augmented before they were used with translation, scale, rotation, brightness, contrast, saturation, and hue. All images were resized to 244×244 and standardized based on the dataset's channel-wise mean and standard deviation.

The model was trained for 30 epochs using the Adam optimizer with a learning rate of 10^{-5} and weight decay of 10^{-4} , with 128 image pairs per batch.

B. Flights

Four flights³ were selected for testing, and they have been used in earlier projects by Mantelli et al. [10] and Stein et al. [11]. All the flights contain satellite images of 4800×4800 pixels taken from different times and a ground truth collected from the GNSS sensor on the UAV. The video feed from the flights has been sampled at 1Hz. All the flights cover a variety of buildings, forest areas, and fields with changing altitudes. A Table of flight details can be seen in Table II, and Fig. 3 shows the trajectories of the flights and the VO data plotted.

Flight 1, used by both Mantelli et al. [10] and Stein et al. [11], and flight 2, used by Mantelli et al. [10], are using the same satellite images over the Brazilian city Arroio do Meio (AdM). The images cover approximately 1.09 km^2 , and five satellite images were taken between 2010 and 2015. While using the same satellite maps, the two flights are over different areas, shown by Fig. 3(a) and (b). Both flights used the same UAV with a gimbal and an altitude change of around 100 meters within the flights.

²<https://simplemaps.com/data/br-cities>

³Dataset available at <https://doi.org/10.5281/zenodo.13748357>

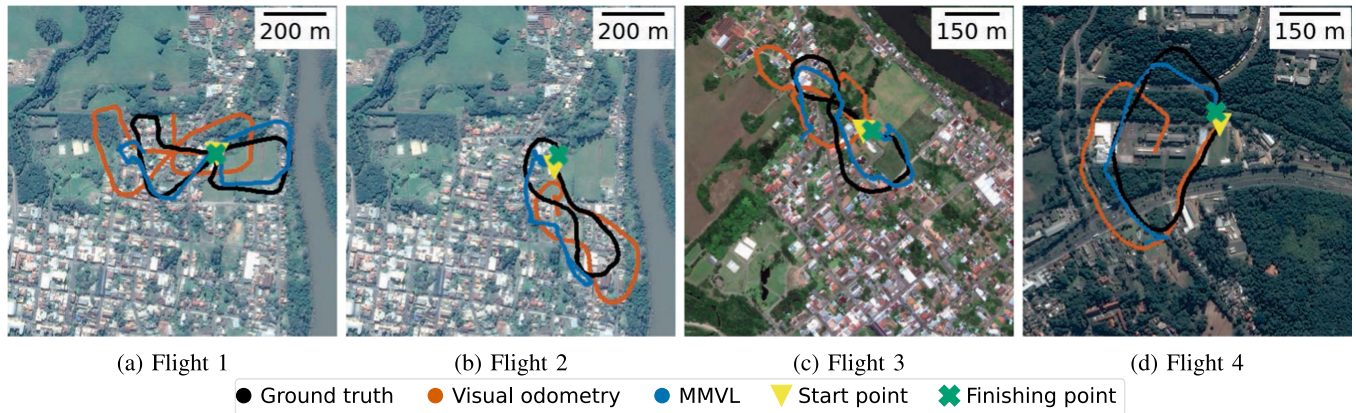


Fig. 3. Trajectories of the four flights together with the VO estimated trajectories used as input for the global localization process. The estimated trajectory after convergence from MMVL is shown in blue.

TABLE II
FLIGHT DETAILS

	Flight 1	Flight 2	Flight 3	Flight 4
Gimbal	Yes	Yes	No	Yes
Length (m)	1800	1340	1120	1090
Map size (m^2)	1.09	1.09	0.81	0.7
Drone image (p)	1280x720	1280x720	1247x917	1280x720
Altitude (m)	35-130	40-145	34-180	45-70
Images	358	269	291	156
Map variations	5	5	2	5
VO drift (m)	187	178	221	187
MAE VO (m)	116	103	111	79

Flight 3, used only by Stein et al. [11], uses two different satellite images over the same city, AdM, taken from 2010 and 2018. Flight 3 used a drone and camera different from Flight 1 and 2 without a gimbal and had the worst VO drift with 221 meters drift by the end.

Flight 4 is over one of the campuses of Universidade Federal do Rio Grande do Sul. It was used by Mantelli et al. [10] and differs significantly from the rest of the flights. The buildings are more sparse than the previous ones, and there are more and larger forest areas with some roads going through. The five available maps were taken from 2013 to 2017. This is also the shortest flight, lasting 156 seconds.

C. Model Accuracy

A script for the similarity model was developed to test its accuracy. The model is trained using satellite images only, but the inference is with UAV and satellite images. The domain shift will impact the final accuracy of the model. Every UAV image in flight 1 was compared to the patch from the correct location of the UAV, called *positive pairs*. Scores above 0.5 were classified as correct. Each image was also compared with ten random places with no overlap of any of the images, called *negative pairs*. Since it was expected to have a low score on these images, scores under 0.5 were classified as correct. The final scores were split into positive and negative pairs and separated for every satellite map.

Fig. 4 shows the scores on the different maps of flight 1. The results show that the model is generally relatively stable, with

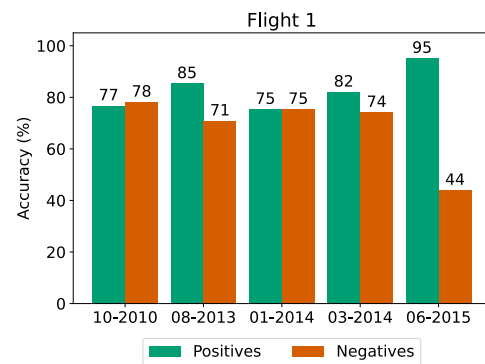


Fig. 4. Model accuracy of positive and negative image pairs tested on the five maps of flight 1.

the difference between positive and negative pairs being small for 01-2014, 10-2010 and 03-2014. The map from 08-2013 has a slightly more significant difference, and with the map from 06-2015, the difference is big with almost 50 percent points. While this test is not a direct measure of the model's performance on localization, it shows that different versions of satellite images can impact the final result. It shows the difficulty of using satellite images as reference maps, especially when training on different scales.

V. EXPERIMENTS AND RESULTS

This section will test and discuss the results of MMVL on real-world flights. Then MMVL will be compared to two other methods, abBRIEF and SIVL, before ending this section with a brief discussion about the computation time of the systems.

A. Localization Results

The results for running MMVL on the four flights can be seen in Table III. The Table shows Mean average error (MAE), MAE after convergence, time to converge, final error, and percentage of converged runs. Convergence is defined as the first occurrence where the error is below 100 meters for ten consecutive iterations. The results are averages of 20 runs, running with

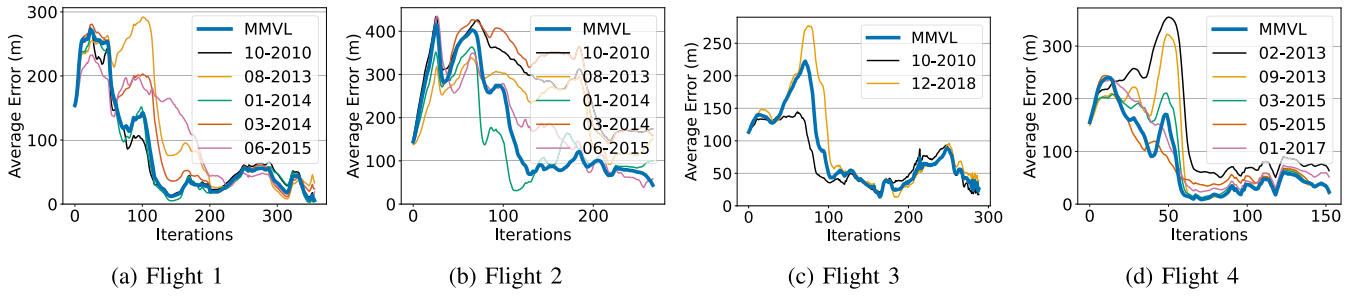


Fig. 5. Error of the estimated localization compared with the ground truth. Showing the average of 20 of each satellite map variation and the combination of all of them.

TABLE III
COMPARISONS WITH SIVL[#] AND abBRIEF

		MMVL	SIVL [#]	abBRIEF ^{**}
Flight 1	MAE (m)	81.0	119.3	66.0
	Time to conv. (iter)	96.8	17.7*	22.6*
	Conv. rate	1.0	0.96	0.9
	MAE conv. (m)	36.2	102.6*	17.8*
	Final error (m)	5.7	180.6	92.8
	Final conv. rate	1.0	0.58	0.68
Flight 2	MAE	176.7	329.0	407.7
	Time to conv. (iter)	134.4	13.0*	80.1*
	Conv. rate	1.0	0.46	0.45
	MAE conv. (m)	71.3	289.5*	332.0*
	Final error (m)	43.8	316.9	329.0
	Final conv. rate	1.0	0.02	0.04
Flight 3	MAE (m)	81.5	359.0	365.1
	Time to conv. (iter)	88.6	34.2*	92.8*
	Conv. rate	1.0	0.45	0.2
	MAE conv. (m)	47.6	472.0	264.6*
	Final error (m)	25.9	399.0	403.8
	Final conv. rate	1.0	0.0	0.05
Flight 4	MAE (m)	80.3	45.9	156.9
	Time to conv. (iter)	49.3	22.8	27.8
	Conv. rate	1.0	1.0	1.0
	MAE conv. (m)	33.5	29.8	163.2
	Final error (m)	23.0	62.7	267.7
	Final conv. rate	1.0	1.0	0.0

* Calculated from the runs that did converge, the rest is removed. ** abBRIEF was run with 50000 particles

10000 particles in the MCL. In addition, Fig. 5 shows the average error over time and an example of an estimated trajectory after convergence is shown in Fig. 3. MMVL have been run using all available satellite maps simultaneously and compared with the system using the maps individually.

MMVL has a long convergence time on flight 1, with 96.8 iterations. After this, the average error is low, at 36.2 meters. Fig. 5(a) shows that all configurations converge and achieve a similar accuracy at the end. However, three configurations of the individual maps, 08-2013, 03-2014, 06-2015, take longer to converge, while MMVL has a similar convergence time to the best individual maps.

The results for flight 2 show the main benefit of the proposed method. Fig. 5(b) shows that most runs with individual maps struggle. 03-2014 and 10-2010 never converge to the correct point, and 08-2013 briefly have an error below 100 meters, but increases after only a few iterations. Of the two maps that converge, 06-2015 takes a long time until convergence. Even with most maps not achieving correct convergence, running

them all in combination with MMVL does. The MAE is higher than flight 1, with a 71.3-meter error after convergence, and the convergence time is longer with 134 iterations. However, MMVL always retains localization, and the end error is lower with only 37.2 meters.

For flight 3, Fig. 5(c), and flight 4, Fig. 5(d), the results are similar. In both flights, every run with individual satellite maps converges and works, and the same is true for MMVL. MMVL uses 88.6 iterations to converge in flight 3 but has a period at around iteration 250 where the error grows. This gives flight 3 a MAE after convergence of 47.6 meters. In flight 4, the error is more stable, and MMVL follows the best individual map, making the MAE after convergence 33.5 meters.

Based on Fig. 5, it can seem like the localization error of MMVL is a bit unstable. This is a common problem in AVL. It is often due to homogeneous regions such as forests and fields, which decreases the accuracy of the similarity model and VO. For MMVL, it is visible in every flight, but especially in flight 3, where the last part is over a field with very similar colors and details (Fig. 3(c)). This causes the error, which is below 50 meters between 100 and 200 iterations, to increase significantly for a period.

The results show that MMVL works and converges correctly in every flight using all the available maps. This is even the case when several maps do not work individually. In addition, the accuracy of running multiple maps is similar to that of the best maps on the flights. Therefore, using a combination of maps does not seem to decrease accuracy. Despite taking relatively long to converge with 49-134 iterations, MMVL achieves high robustness and reliability, providing accurate results in all experiments. In order to gain an understanding of the performance, MMVL was compared with other proposals.

B. Comparison

Two models were chosen to compare with MMVL. Both use MCL and satellite images. The first method, abBRIEF, was proposed by Mantelli et al. [10]. It is a popular baseline for AVL, and the implementation was available for this letter. AbBRIEF locates the UAV in four dimensions, (x, y, z, θ) , but for the comparison with MMVL, only (x, y) is used.

The second method is a slightly modified version of Kinnari et al.'s proposed SIVL [13]. The CNN was trained on our dataset but in the same way as Kinnari et al. The model was used in the MCL algorithm implemented for MMVL without the

TABLE IV
COMPUTATION TIME

	MMVL-1	MMVL-2	MMVL-5	MMVL-1-nc	SIVL [#]	abBRIEF
Caching (ms)	40889	80375	220194	0	0	0
Motion update (ms)	191	190	192	189	189	-
Sensor update (ms)	64	101	161	225355	66823	-
Resample (ms)	95	94	95	90	98	-
Total online (ms)	350	385	448	225634	67110	3850

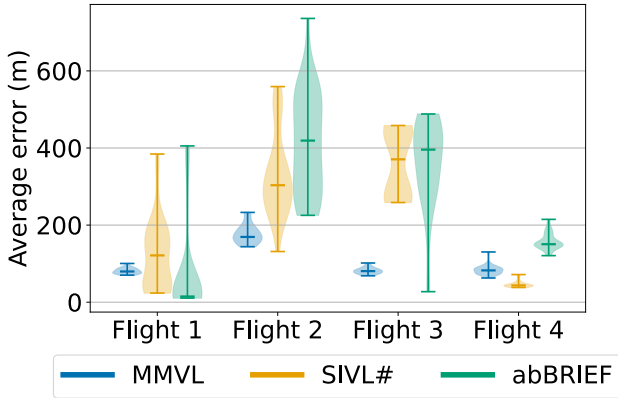


Fig. 6. Violin plot of the average error for all the flights.

orthorectification process from the original proposal. The main reason for not including this was that the code was not publicly available at the time of writing. The process was implemented to use images taken from an angle, however all flights in this work use approximately orthogonal images already. In addition, the orthorectification process removes camera distortions and the altitude variable, and can possibly impact the accuracy. Due to the modifications, our implementation will be noted with a mark, SIVL[#].

The two methods were run ten times on each satellite image. The results are combined to show the expected results when choosing a random map. Like MMVL, SIVL[#] was run with 10000 particles, while abBRIEF was run with 50000 according to suggested values by Mantelli et al. [10]. Table III shows the results after running SIVL[#] and abBRIEF with the four flights, and Fig. 6 shows the average error of all the runs.

The results of SIVL[#] show a rapid convergence rate of the converged runs. On flight 1, 96% of all runs converge with an average of only 17.7 iterations, much quicker than the 96.8 iterations that MMVL used. However, over time, SIVL[#] cannot keep the correct location, and in the end, only 58% of the runs had an error below 100 meters. Fig. 6 shows that in the best cases, SIVL[#] have a lower average error than MMVL, but the variance is considerable, and the mean of errors is larger than MMVL. For flights 2 and 3, it converges quicker than MMVL, but only a very few runs converge at all, and in the end, only 2% and 0% of the runs converge. This means a much higher MAE than MMVL, even on flight 2, where MMVL did have a relatively high error of 176.7 meters, SIVL[#] had a higher MAE of 329.0 meters. In flight 4, SIVL[#] worked perfectly, out-competing MMVL in almost every metric with a MAE of only 45.9 meters versus 80.3 meters for MMVL. It is worth mentioning that a difference between the performance of our implementation of SIVL[#] and the original proposal can be noticed. There are likely

several reasons for this, including a more challenging situation with real VO and multiple satellite maps. In addition, it is not unlikely that the removal of the orthorectification process also impacted the results.

AbBRIEF performs well on flight 1 but struggles on the other flights. On flight 1, it is quicker to converge than MMVL and has a lower error after convergence with only a 17.8-meter error versus 36.2 meters for MMVL. 90% of the runs converge, but at the end, only 68% have the correct location. Still, MAE is lower for abBRIEF with 66.0 meters versus 81.0 meters for MMVL. For the rest of the flights, MMVL outperforms abBRIEF. As Fig. 6 shows, the average error is high apart from a few runs on flight 3. For flights 2 and 3, the variance is considerable as well. In flight 4, the variance shown by Fig. 6 is much lower, but still, the MAE is extensive, with 156.9 meters versus 80.3 meters for MMVL.

The comparisons give a clearer view of the characteristics of MMVL. In a few cases, it is not performing better than the competing methods, but overall when considering all flights, MMVL is an improvement. MMVL converges slower and has a relatively high error after converging, compared to the successful runs of the competing methods, which are much less often. However, Fig. 6 shows that the performance of the SIVL[#] and abBRIEF varies significantly on the flights and maps used. MMVL is generally less impacted by different runs and flights.

C. Computation Time

To be able to run online on a UAV with limited resources, computation efficiency is essential for the localization methods. Table IV shows computation time results of MMVL and other approaches. The number behind MMVL denotes the number of maps being used. MMVL is compared with MMVL using one map and no caching, denoted as MMVL-1-nc and with abBRIEF and SIVL[#]. The tests were run on AMD Ryzen 5 5600 G with 66 GB memory. It is worth noting that the implementation is not optimized for speed, and the computer is more powerful than what would be available on UAVs. However, the results still show the general computation requirements for the different systems and make it possible to compare them.

The results are clear. Using ResNet34 without caching is not feasible, MMVL-1-nc using almost 4 minutes, or 645 times longer than the equivalent model using caching. Caching requires some preparation before the flight. In the case of MMVL-1, it is about 40 seconds. It is still much lower than just one iteration of MMVL-1-nc.

Using multiple satellite maps simultaneously impacts the time, but the increase is minor. The offline preparation increases as a multiple to the number of maps, but the online process can benefit from only processing the UAV image once per

iteration. The total increase is not more than 28%, going from 350 milliseconds when using one map to 448 milliseconds with 5 maps. This means using multiple satellite images in MMVL is a viable and efficient option.

MMVL is also more efficient than the two competing methods. SIVL[#] uses over a minute per iteration, showing that caching is needed even with a small 6-layer CNN. abBRIEF is quicker but still uses almost 9 times longer than MMVL with five maps.

VI. CONCLUSION

This letter proposes a novel approach to visual localization called MMVL. It uses a CNN model to compare images with a cache to speed up the compute time. The comparisons are used to create a probability map showing the most likely locations of the UAV of the specific timestep. The probability maps from different satellite maps are combined using a weighted average, where the weights are from the normalization of each probability map. The probability map is then used in a MCL algorithm to estimate the correct location. With MMVL, multiple satellite maps can be combined simultaneously with a weighted average while calculating the probability map.

MMVL was tested on four flights with changing altitude, satellite maps, and camera setup. The results show that MMVL is a clear improvement in some areas but not in every. MMVL have a slower convergence time and a higher error than the other methods, SIVL[#] and abBRIEF, in the runs where they are converging correctly. However, using multiple satellite maps simultaneously does not seem to impact the error, as the system achieves the same accuracy as the best results when running individual satellite maps.

Where MMVL is a big improvement is the system's robustness. MMVL converged in every run achieving 100% convergence rate. Compared with the two other methods, this is a clear improvement. The other methods performed with a convergence rate between 0-100% depending on the flights. Being robust to different flights was the main goal of the proposal. MMVL is also computationally efficient and extensively outperforms the two other methods. This shows that using multiple satellite maps can be a good strategy.

For future work, using multiple satellite maps enables research on satellite maps and drone flights with more significant discrepancies like snow and rain. In addition, it is possible to use different CNN models trained using different methods in the same way multiple satellite maps are used. This can further increase the robustness and possibly accuracy.

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